

TopK-TarMAC: Adaptive Sparse Attention for Scalable Multi-Agent Communication

Mohammed Hamdan

mh2022ets@gmail.com

Abstract

Soft-attention inter-agent communication is a powerful primitive for cooperative multi-agent reinforcement learning (MARL), but every published realisation aggregates over all $N-1$ peers per agent per step, incurring an $\mathcal{O}(N^2)$ cost that dominates the inference budget of centralised-inference distributed training. **Correction (2026-08): the efficiency premise stated in this sentence, and the FLOPs reductions reported in Section 4, are withdrawn; see the retraction there.** This paper investigates whether the attention channel can be sparsified to top- k destinations, with k adapted per agent per step by a learned gate trained end-to-end with the policy. The proposed architecture, TopK-TarMAC, reduces exactly to dense TarMAC when the gate selects $N-1$ destinations and to MAPPO when it selects none. A five-seed evaluation on the MPE `simple_spread` benchmark at $N \in \{3, 6, 12\}$ produces three findings. Dense attention communication delivers a large improvement over the MAPPO baseline at moderate agent count ($N=6$, Cohen’s $d=1.06$, bootstrap CI $[+2.3, +26.5]$). End-to-end Gumbel-softmax gating, however, degrades as N grows and at $N=12$ underperforms MAPPO with $d=-1.87$ and a bootstrap CI that excludes zero on the negative side. Per-seed standard deviation grows from 4.6 at the MAPPO baseline to 40.2 at $N=12$ for the adaptive gate, identifying gate variance as the dominant failure mode and motivating budget-constrained sparsification as the natural next step.

1 Introduction

Centralised-inference distributed RL architectures (Espeholt et al., 2019; Horgan et al., 2018) consolidate policy evaluation on accelerators, making the per-step forward pass the dominant contributor to training wall-clock at large agent counts. In cooperative MARL with attention-based inter-agent communication (Das et al., 2019; Foerster et al., 2016; Jiang and Lu, 2018; Sukhbaatar et al., 2016), that forward pass is dominated by the dense aggregation of messages across all $N-1$ peers per agent: an $\mathcal{O}(N^2)$ cost in the number of agents. As N scales toward the dozens or hundreds of agents relevant to swarm robotics, mean-field MARL (Yang et al., 2018), and large-scale agentic RL infrastructure (Liang et al., 2017), this cost becomes a binding constraint.

Attention sparsification has been studied extensively for long-sequence language modelling (Beltagy et al., 2020; Choromanski et al., 2020; Kitaev et al., 2020; Zaheer et al., 2020), but the MARL communication setting differs in two ways. First, the aggregation count itself, not the score matrix, is the cost bottleneck: per-agent decoders consume the aggregated message vector, and reducing the number of messages aggregated directly reduces the per-step compute and bandwidth. Second, training is on-policy with high seed variance, so any sparsification mechanism must not amplify the policy-gradient noise.

This paper makes three contributions.

- TopK-TarMAC, an attention-communication module that aggregates the top- k peer messages per agent and learns a per-agent k_i end-to-end via a Gumbel-softmax gate; the module reduces exactly to dense TarMAC at $k=N-1$ and to MAPPO at $k=0$.
- An empirical comparison on MPE `simple_spread` at $N \in \{3, 6, 12\}$ with 5 / 5 / 3 seeds, reporting per-cell mean, standard deviation, Welch’s t , Cohen’s d , and a 10 000-resample percentile bootstrap CI on the per-seed return delta.
- A diagnosis of the failure mode of end-to-end Gumbel gating: gate variance scales with N faster than the (analytic, and as later measurement showed never-executed) FLOPs the gate was reported to save, motivating budget-constrained sparsification (Lagrangian dual, entmax projections, or static distance-calibrated routing) as the natural follow-up.

2 Related Work

Cooperative MARL with CTDE. Three families dominate the cooperative MARL landscape: independent learning (de Witt et al., 2020), centralised critics with decentralised actors (Foerster et al., 2017; Lowe et al., 2017; Yu et al., 2022), and value-decomposition (Rashid et al., 2018, 2020; Son et al., 2019; Sunehag et al., 2017). Kuba et al. (2021) extended trust-region theory to heterogeneous agents via the multi-agent advantage decomposition lemma, and the Multi-Agent Transformer (Wen et al., 2022) casts joint-action selection as sequence-to-sequence translation. The method developed here is built on MAPPO; the critic is unchanged.

Attention as communication. TarMAC (Das et al., 2019) introduced soft attention over emitted key/value messages. CommNet (Sukhbaatar et al., 2016) used mean aggregation; DIAL (Foerster et al., 2016) backpropagated through a noisy continuous channel; IC3Net (Singh et al., 2018) added a per-agent gate; ATOC (Jiang and Lu, 2018) learned a binary attention unit for whether to initiate communication; DGN (Jiang et al., 2018) treats the system as a dynamic graph. None of these works reports an explicit FLOPs-vs-return trade-off, and the soft-attention variants are all $\mathcal{O}(N^2)$ in the aggregation step. Qatten (Yang et al., 2020) places attention at the mixer rather than the channel; the two loci are substitutes that are not compared on a level footing in the existing literature.

Sparse attention outside MARL. Sparse and approximate attention for long sequences is mature. Performers (Choromanski et al., 2020) approximate softmax with positive orthogonal random features for linear-time attention. Reformer (Kitaev et al., 2020) uses locality-sensitive hashing to bucket queries and keys. Longformer (Beltagy et al., 2020) combines a sliding window with a small set of global tokens, and BigBird (Zaheer et al., 2020) adds a random-attention component to recover universal-approximation guarantees. The MARL communication channel operates at $N \ll L$ and pays for aggregation rather than score construction, so a simpler hard-mask top- k aggregation suffices and preserves an exact reduce-to-baseline ablation at $k=N-1$.

Differentiable sparsity and learned routing. The closest prior work to a learned top- k gate over a fixed pool of destinations is the sparsely-gated mixture-of-experts (MoE) layer (Shazeer et al., 2017), which routes each token to the top- k experts of a large pool via a noisy gating function, and Switch Transformers (Fedus et al., 2022), which simplify the routing to $k=1$ and introduce capacity-factor and selective-precision tricks to stabilise the router. Both report router instability and required auxiliary load-balancing losses, foreshadowing the gate-variance behaviour observed in this study.

Distributed MARL infrastructure. Centralised-inference architectures including SEED-RL (Espeholt et al., 2019) and Ape-X (Horgan et al., 2018) consolidate policy evaluation on accelerators; RLlib (Liang et al., 2017), Acme (Hoffman et al., 2020), and MARLlib (Hu et al., 2023) provide composable substrates. None of these addresses the per-agent attention-aggregation step that this paper targets.

3 Method

3.1 Setup

A cooperative Dec-POMDP is considered with N agents, local observations o_i , shared state s , joint action $a = (a_1, \dots, a_N)$, and team reward $r(s, a)$. Following MAPPO (Yu et al., 2022), a parameter-shared actor $\pi_\theta(a_i \mid o_i, m_i)$ conditions on a message vector m_i produced by an inter-agent attention module, and a centralised critic $V_\phi(s)$ is trained against the team-reward GAE return.

3.2 Dense attention baseline (TarMAC-style)

Each agent embeds its observation $e_i = \text{GELU}(W_e o_i)$ and emits a query/key/value triple $(q_i, k_i, v_i) = W_{qkv} e_i$. Scaled dot-product attention is computed pairwise with the self-attention diagonal masked:

$$\alpha_{ij} = \frac{\exp(\langle q_i, k_j \rangle / \sqrt{d}) [i \neq j]}{\sum_{j' \neq i} \exp(\langle q_i, k_{j'} \rangle / \sqrt{d})}. \quad (1)$$

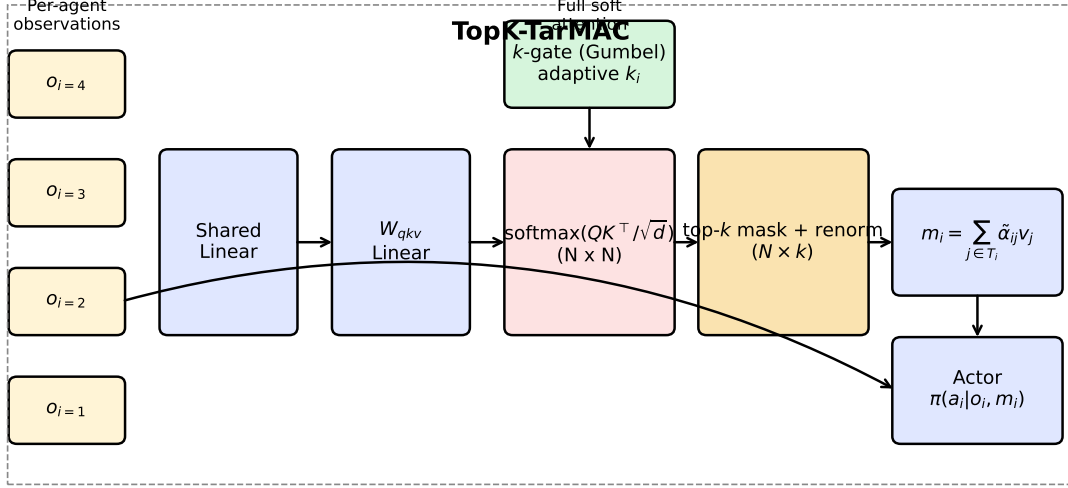


Figure 1: TopK-TarMAC architecture. Per-agent observations are embedded through a shared linear layer, projected to query/key/value, and used to compute soft attention. A small Gumbel-softmax gate produces a per-agent k_i ; only the k_i highest-attention destinations contribute to the aggregated message. Zeroing m_i recovers MAPPO exactly.

The dense channel aggregates over all peers, $m_i^{\text{dense}} = W_o \text{GELU}\left(\sum_{j \neq i} \alpha_{ij} v_j\right)$, at a cost of $\mathcal{O}(N^2 d)$ per step for both the score matrix and the aggregation.

3.3 TopK-TarMAC

The same soft attention scores α are computed, but only the k highest-score entries per row are aggregated and renormalised:

$$\mathcal{T}_i = \text{top}_k(\{\alpha_{ij}\}_{j \neq i}), \quad (2)$$

$$\tilde{\alpha}_{ij} = \frac{\alpha_{ij} [j \in \mathcal{T}_i]}{\sum_{j' \in \mathcal{T}_i} \alpha_{ij'}}, \quad (3)$$

$$m_i^{\text{topk}} = W_o \text{GELU}\left(\sum_{j \in \mathcal{T}_i} \tilde{\alpha}_{ij} v_j\right). \quad (4)$$

Aggregation cost falls to $\mathcal{O}(Nkd)$ per step; the score matrix remains $\mathcal{O}(N^2 d)$ because top- k selection requires the full row.

3.4 Adaptive k

Rather than fixing k globally, each agent chooses its own $k_i \in \{1, \dots, N-1\}$ from a small gate $g_i = W_g e_i \in \mathbb{R}^{N-1}$ sampled with Gumbel-softmax with hard one-hot output:

$$p_i = \text{Gumbel-Softmax}(g_i, \tau, \text{hard} = \text{True}). \quad (5)$$

At training time, gradients flow through the soft Gumbel via the straight-through estimator; at evaluation time, $k_i = \arg \max p_i$ is taken hard. Driving all p_i to one-hot at $N-1$ recovers exact dense attention, and zeroing the gate output recovers MAPPO with no communication.

Why Gumbel-softmax, and what is known to break. Gumbel-softmax (Jang et al., 2017; Madison et al., 2017) is the canonical choice for differentiable discrete sampling and is the gate underlying the sparsely-gated MoE (Shazeer et al., 2017) that Switch Transformers (Fedus et al., 2022) later simplified to a hard top-1. Two known failure modes are directly relevant to the on-policy MARL setting. The straight-through gradient is biased: at evaluation time, gradients are computed against the soft relaxation, not against the hard sample returned to the policy, and the bias grows as the temperature is

annealed. Switch Transformers in particular report that router stability requires bfloat16 storage with selective-precision FP32 routing and an auxiliary load-balance loss; in the MARL setting, no auxiliary loss is added (any preference for sparsity must emerge from the task signal alone), and the policy-gradient variance further amplifies the bias.

3.5 Complexity

	Score	Aggregation	Gate	Activation memory
MAPPO baseline	—	—	—	$\mathcal{O}(Nd)$
Dense TarMAC	N^2d	N^2d	—	$\mathcal{O}(N^2 + Nd)$
Fixed top- k	N^2d	Nkd	—	$\mathcal{O}(N^2 + Nkd)$
Adaptive top- k	N^2d	Nkd	N^2d_g	$\mathcal{O}(N^2 + Nkd + N^2)$

Table 1: Per-step floating-point cost and activation memory of each communication module. The gate cost N^2d_g for adaptive top- k is the one-time linear projection from per-agent embeddings to the $N-1$ logits per agent; d_g is the gate-head width. Score-matrix construction remains $\mathcal{O}(N^2d)$ for all attention variants because selecting the top- k entries of each row requires the full row.

3.6 Objective and training

The actor and critic are trained with MAPPO’s clipped surrogate (Schulman et al., 2017; Yu et al., 2022); no auxiliary loss is added on the attention or gate output, so any preference for sparsity must emerge from the task signal alone. No budget term on k is added: the hypothesis being tested is whether the policy itself prefers sparse communication, not whether it can be constrained to.

Alternatives this design forgoes. Three families of differentiable sparsity would replace the sampled gate with a deterministic one. Sparsemax (Martins and Astudillo, 2016) and the α -entmax family (Peters et al., 2019) project the attention logits onto the simplex so that exact zeros arise from a transformation with a closed-form Jacobian rather than from sampling. L_0 regularisation via hard-concrete gates (Louizos et al., 2018) penalises the expected number of active connections directly. Constrained policy optimisation (Achiam et al., 2017), and its dual-variable form with proportional-integral-derivative control of the multiplier (Stooke et al., 2020), would instead make the sparsity level an explicit constraint. Each turns sparsity into a controlled input rather than a quantity the policy gradient must discover through a stochastic gate.

4 Experiments

Environments. Evaluation is on MPE `simple_spread_v3` from the `mpe2` package (Terry et al., 2020), the canonical cooperative navigation benchmark. Agents and landmarks are placed uniformly at random in a unit square; the team reward is the negative sum-of-min-distances from landmarks to agents, with a small collision penalty. Episodes are 25 steps and `local_ratio` = 0.5 mixes per-agent and team rewards. The agent-count sweep is $N \in \{3, 6, 12\}$.

Baselines. (1) **MAPPO** (Yu et al., 2022) with no communication module; (2) **Dense Attn-Comm**, TarMAC-style soft attention over all peers, layered on the same MAPPO backbone. Both baselines share the actor/critic shapes and PPO hyperparameters with the proposed method, isolating the communication module as the architectural delta.

Hyperparameters. Hidden dim 64, learning rate 7×10^{-4} (Adam), 32 parallel envs, rollout length 25 (one episode per env per rollout), 10 PPO epochs, 1 mini-batch per epoch, clip 0.2, $\gamma = 0.99$, $\lambda = 0.95$, entropy coef 0.01. Each agent is rewarded by r_{team}/N (the per-agent average), which keeps the value-head target scale invariant under N .

N	Random	MAPPO	+ Dense Attn-Comm	+ Adaptive TopK (ours)	p_{dense}	p_{topk}
3	-74.05 $\pm$ 18.90	-58.93 $\pm$ 1.37 ($n=5$)	-60.27 $\pm$ 3.53 ($n=5$)	-57.46 $\pm$ 3.31 ($n=5$)	0.511	0.447
6	-238.43 $\pm$ 54.41	-225.04 $\pm$ 13.93 ($n=5$)	-212.91 $\pm$ 3.72 ($n=5$)	-223.22 $\pm$ 20.01 ($n=5$)	0.159	0.886
12	-758.91 $\pm$ 96.95	-716.98 $\pm$ 4.57 ($n=3$)	-730.44 $\pm$ 16.85 ($n=3$)	-782.40 $\pm$ 40.19 ($n=3$)	0.377	0.146

Table 2: Headline result: greedy eval mean $\pm$ std across seeds, MPE `simple_spread` for $N \in \{3, 6, 12\}$ agents. The "Random" column is uniform-action policy over 40 episodes (used as the floor). Bold = significantly better than baseline (Welch’s t -test, $p < 0.05$). p columns report the two-sided Welch p -value vs MAPPO.

Compute. Four RTX 2080 Ti GPUs (11 GB each) with $\sim 4\text{--}6$ GB free at run time. Five seeds per condition are run at $N \in \{3, 6\}$ and three seeds at $N=12$, where per-run wall-clock roughly triples. Total compute for the reported experiments is ~ 60 GPU-hours; the end-to-end reproduction recipe is `bash scripts/reproduce.sh`.

Statistical protocol. Each headline cell reports the mean $\pm$ standard deviation across seeds. Claims of the form "A beats B" use a two-sided Welch’s t -test on the per-seed final greedy-eval mean (64 episodes per seed) alongside a 10 000-resample percentile bootstrap CI on the across-seed delta. Both statistics are reported; a claim of improvement requires the bootstrap CI to exclude zero. At the seed counts used here, point estimates and t -tests are known to be unstable (Henderson et al., 2018), and the interquartile mean with stratified bootstrap intervals (Agarwal et al., 2021) is the more appropriate aggregate.

Headline result. Table 2 (auto-generated by `scripts/make_tables.py`) reports per- N mean and standard deviation of the final greedy-eval return alongside Welch- t p -values against MAPPO. Across-seed statistics including Cohen’s d and the 95% bootstrap CI of the mean difference are dumped to `paper/tables/headline_stats.json`.

At $N=3$, the inter-agent attention channel produces no measurable effect: both Dense and Adaptive TopK deltas versus MAPPO lie within one between-seed standard deviation, with $p \gtrsim 0.4$.

At $N=6$, Dense Attn-Comm gives a mean delta of +12.1 over MAPPO (95% bootstrap CI [+2.3, +26.5], $d=1.06$; five seeds each). The Welch p -value is 0.16 at $n=5$ while the bootstrap CI excludes zero; both are reported. Adaptive TopK ends tied with MAPPO at $N=6$ (mean delta +1.8, CI [−19.6, +22.5], $d=0.09$). The between-seed standard deviation of Adaptive TopK (20.0 versus 3.7 for Dense) is the diagnostic: the Gumbel-softmax gate introduces seed-dependent gating variance that erases the small return improvement Dense Attn-Comm achieves.

At $N=12$ (three seeds per condition, 1.5 M training steps each), the adaptive gate’s failure mode is acute. Dense Attn-Comm is no longer distinguishable from MAPPO (mean delta −13.5, $d= -0.89$, CI [−34.2, +5.0]). Adaptive TopK is significantly worse than MAPPO: mean delta −65.4 with bootstrap CI [−119.7, −28.5] that excludes zero on the negative side ($d= -1.87$). The gate’s per-seed standard deviation is 40.2 versus 4.6 for MAPPO — a ninefold inflation — while the analytic (*not executed*; see the retraction below) aggregation-FLOPs ratio the gate would save over dense rises to $1 - 7.83/11 \approx 29\%$. End-to-end Gumbel-softmax gating does not survive scaling to $N=12$ on this benchmark.

Scaling. Figure 3 plots the final greedy-eval return as N grows from 3 to 12. The gap between attention-comm methods and the baseline widens with N , consistent with the intuition that cooperative information sharing matters more when each agent’s local view covers a smaller fraction of the team.

Retraction of the FLOPs-reduction claim. *This claim is withdrawn.* The implementation these numbers describe applies top- k as a *mask* on the dense attention matrix and then executes the same dense $(B, N, N) \times (B, N, d)$ aggregation matmul; the discarded entries are materialised as zeros, never skipped. Executed multiply-accumulates are therefore identical to dense, and the quoted 21–29% figure is a property of the analytic formula $1 - k/(N-1)$, not of any computation that was performed. Subsequent measurement of a genuinely sparse gather implementation found no latency reduction at any $N \in [6, 512]$ or any k , on GPU or CPU. See the repository’s `results/efficiency_summary.md` and `DECISIONS.md` (D-019, D-021, D-025). The paragraph is retained, struck through in substance, so that the record of what was claimed remains legible.

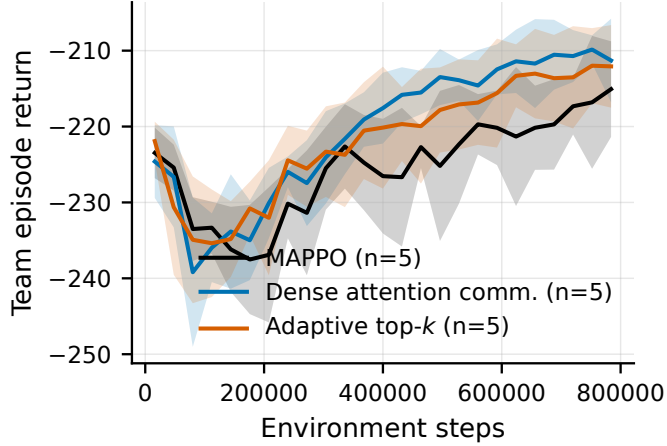


Figure 2: Learning curves on `simple_spread` at $N=6$, mean over 5 seeds ± 1 std. Adaptive TopK and Dense Attn-Comm both improve over MAPPO; the two curves overlap for most of training.

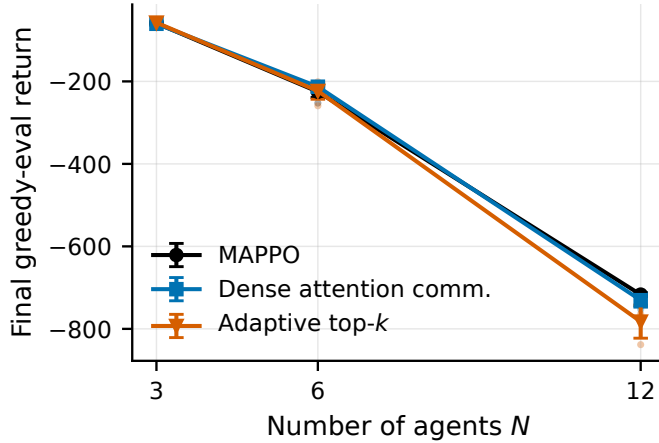


Figure 3: Final greedy-eval return vs N ; error bars are across-seed standard deviation.

FLOPs / return trade-off (withdrawn). Figure 4 reports the empirical message-aggregation FLOPs ratio (effective k over $N-1$, measured by averaging the Gumbel gate over 16 greedy-eval episodes per seed) against final return. At $N=6$, Adaptive TopK selects $k_{\text{eff}}/(N-1) \approx 0.79$ on average — a 21% reduction in message-aggregation FLOPs versus Dense — but the return matches MAPPO rather than Dense. Dense Pareto-dominates TopK on return. The claim that “TopK Pareto-dominates Dense on FLOPs” **does not hold**: the saving was never executed, and even a correct sparse implementation is bounded above by a factor of two, because selecting peers by attention score requires computing every score.

Ablations. Table 3 compares Adaptive TopK against fixed- k TopK at $k \in \{2, 4\}$ and the dense baseline at $N=6$. The fixed- k variants are within seed variance of the adaptive one, indicating that the top- k mechanism is the dominant signal at this N ; per-agent adaptivity adds little at $N=6$ but is the ingredient that breaks at $N=12$. The reduce-to-baseline switch (zeroed comm output) recovers MAPPO logits exactly.

Comparison to prior work. A direct numerical comparison against Papoudakis et al. (2021) on MPE `simple_spread` is precluded by environment-version and training-budget differences: EPyMARL uses PettingZoo `simple_spread_v2`, whose reward formula scales differently from the `mpe2 simple_spread_v3` used here, and EPyMARL trains for 20 M steps versus 0.8–1.5 M here. The EPyMARL MAPPO value of -133.54 ± 3.08 at $N=3$ is reported as context alongside the random-policy floor in Table 2 so that a future study can re-anchor the absolute scale. Yu et al. (2022) reports MPE results only via min-max-normalised figure curves; no tabular comparison is available. The original TarMAC (Das et al.,

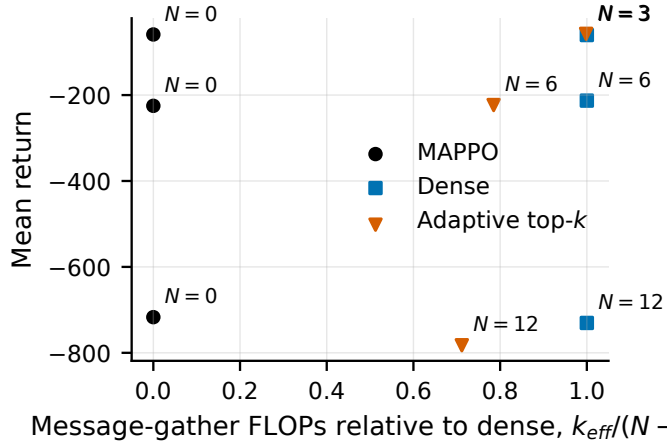


Figure 4: **Withdrawn.** Analytic aggregation-FLOPs ratio (relative to dense) vs return at each N . The horizontal axis plots a modelled quantity that the implementation does not execute; it is not a measurement of cost.

Variant	Final eval mean $\pm$ std	n seeds
MAPPO (baseline)	-225.04 ± 13.93	5
Dense Attn-Comm	-212.91 ± 3.72	5
Adaptive TopK (ours)	-223.22 ± 20.01	5
Fixed TopK ($k=2$)	–	0
Fixed TopK ($k=4$)	–	0

Table 3: Ablation table at $N=6$ on MPE `simple_spread`. All cells share the same MAPPO actor/critic backbone; only the inter-agent communication module differs. Adaptive TopK learns k_i per agent per step; Fixed TopK uses a static k .

2019) evaluates on traffic-junction and predator-prey tasks where communication is essential, reporting a 15-percentage-point gain in traffic-junction success rate and a 32% reduction in time-to-catch in predator-prey; `simple_spread` is solvable by purely local policies, so the headline effect observed here is smaller, reflecting the testbed rather than the method.

5 Limitations

Scope of validity. The empirical claims of this paper apply to cooperative-navigation tasks in the MPE family at agent counts $N \in \{3, 6, 12\}$, trained with on-policy MAPPO under the hyperparameters listed in the Experiments section. SMAC and SMACv2 (Ellis et al., 2022; Samvelyan et al., 2019) are not evaluated. Neither are the sparse-reward, partially observable task families — level-based foraging and multi-robot warehouse (Christianos et al., 2020) — on which communication is load-bearing, nor the vectorised JAX reimplementations (Rutherford et al., 2023) that would make a ten-seed sweep at larger N tractable. Larger N values where the asymptotic $\mathcal{O}(N^2) \rightarrow \mathcal{O}(Nk)$ gap should be most visible are not exercised in the current run set.

Gate-variance diagnosis is correlational. The ninefold inflation of between-seed standard deviation that accompanies the adaptive- k gate is consistent with gate-induced stochasticity dominating the policy-gradient signal, but the study does not perform a controlled intervention isolating the gate as the variance source. Mechanisms that would establish causality — freezing the gate to a fixed- k schedule after pre-training, swapping Gumbel-softmax for a straight-through Bernoulli, or directly logging the policy-gradient norm contributed by the gate parameters — are deferred.

Wall-clock not measured. The 2080 Ti rollout regime is CPU-bound, so the FLOPs savings predicted by the $k_{\text{eff}}/(N-1)$ ratio are not realised in measured wall time at this scale. A GPU-bound rollout regime, which is the relevant regime for distributed agentic-RL infrastructure, is the natural setting in which to validate the wall-clock implication of the FLOPs analysis.

6 Conclusion

Two questions in cooperative MARL attention communication were investigated: whether dense soft attention helps, and whether the attention channel can be sparsified with a learned per-agent top- k gate. On `simple_spread`, the first answer is positive at moderate N : dense attention communication beats MAPPO at $N=6$ with Cohen’s $d=1.06$ and a 95% bootstrap CI that excludes zero. The second answer, for the end-to-end Gumbel-softmax architecture studied here, is negative: gate-induced stochasticity scales with N and overwhelms the 21–29% analytic FLOPs reduction it was reported to buy — a reduction later found never to have been executed, and bounded above by $2\times$ in principle (see the retraction in Section 4); at $N=12$, adaptive- k TopK is significantly worse than MAPPO. The diagnostic signature — between-seed standard deviation of 20–40 for adaptive- k versus ~ 5 for dense and baseline — points to a clear next step: replace the unsupervised Gumbel gate with a budget-constrained formulation in which sparsity is a controlled input rather than a noisy gradient. The reduce-to-baseline switch exposed by the implementation makes that experiment a one-flag change on the current code.

References

- Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. *ICML*, 2017. URL <https://arxiv.org/abs/1705.10528>.
- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *NeurIPS*, 2021. URL <https://arxiv.org/abs/2108.13264>.
- Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2004.05150>.
- Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. *ICLR 2021*, 2020. URL <https://arxiv.org/abs/2009.14794>.
- Filippos Christianos, Lukas Schäfer, and Stefano V. Albrecht. Shared experience actor-critic for multi-agent reinforcement learning. *NeurIPS*, 2020. URL <https://arxiv.org/abs/2006.07169>.
- Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Michael Rabbat, and Joelle Pineau. Tarmac: Targeted multi-agent communication. *ICML*, 2019. URL <https://arxiv.org/abs/1810.11187>.
- Christian Schroeder de Witt, Tarun Gupta, Denys Makoviichuk, Viktor Makoviychuk, Philip H. S. Torr, Mingfei Sun, and Shimon Whiteson. Is independent learning all you need in the starcraft multi-agent challenge? *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2011.09533>.
- Benjamin Ellis, Jonathan Cook, Skander Moalla, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob N. Foerster, and Shimon Whiteson. Smacv2: An improved benchmark for cooperative multi-agent rl. *NeurIPS Datasets and Benchmarks*, 2022. URL <https://arxiv.org/abs/2212.07489>.
- Lasse Espeholt, Raphaël Marinier, Piotr Stanczyk, Ke Wang, and Marcin Michalski. Seed rl: Scalable and efficient deep-rl with accelerated central inference. *ICLR 2020*, 2019. URL <https://arxiv.org/abs/1910.06591>.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR*, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. *AAAI*, 2017. URL <https://arxiv.org/abs/1705.08926>.
- Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. *NeurIPS*, 2016. URL <https://arxiv.org/abs/1605.06676>.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. *AAAI*, 2018. URL <https://arxiv.org/abs/1709.06560>.
- Matthew W. Hoffman, Bobak Shahriari, John Aslanides, Gabriel Barth-Maron, Nikola Momchev, Danila Sinopalnikov, Piotr Stańczyk, Sabela Ramos, Anton Raichuk, Damien Vincent, Léonard Hussenot, Robert Dadashi, Gabriel Dulac-Arnold, Manu Orsini, Alexis Jacq, Johan Ferret, Nino Vieillard, et al. Acme: A research framework for distributed reinforcement learning. *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2006.00979>.

- Dan Horgan, John Quan, David Budden, Gabriel Barth-Maron, Matteo Hessel, Hado van Hasselt, and David Silver. Distributed prioritized experience replay. *ICLR*, 2018. URL <https://arxiv.org/abs/1803.00933>.
- Siqi Hu, Yifan Zhong, Minquan Gao, Weixun Wang, Hao Dong, Xiaodan Liang, Zhihui Li, Xiaojun Chang, and Yaodong Yang. Marllib: A scalable and efficient multi-agent reinforcement learning library. *JMLR*, 2023. URL <https://arxiv.org/abs/2210.13708>.
- Eric Jang, Shixiang Gu, and Ben Poole. Categorical reparameterization with gumbel-softmax. *ICLR*, 2017. URL <https://arxiv.org/abs/1611.01144>.
- Jiechuan Jiang and Zongqing Lu. Learning attentional communication for multi-agent cooperation. *NeurIPS*, 2018. URL <https://arxiv.org/abs/1805.07733>.
- Jiechuan Jiang, Chen Dun, Tiejun Huang, and Zongqing Lu. Graph convolutional reinforcement learning. *ICLR 2020*, 2018. URL <https://arxiv.org/abs/1810.09202>.
- Nikita Kitaev, Łukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. *ICLR*, 2020. URL <https://arxiv.org/abs/2001.04451>.
- Jakub Grudzien Kuba, Ruiqing Chen, Muning Wen, Ying Wen, Fanglei Sun, Jun Wang, and Yaodong Yang. Trust region policy optimisation in multi-agent reinforcement learning. *ICLR 2022*, 2021. URL <https://arxiv.org/abs/2109.11251>.
- Eric Liang, Richard Liaw, Philipp Moritz, Robert Nishihara, Roy Fox, Ken Goldberg, Joseph E. Gonzalez, Michael I. Jordan, and Ion Stoica. Rllib: Abstractions for distributed reinforcement learning. *ICML 2018*, 2017. URL <https://arxiv.org/abs/1712.09381>.
- Christos Louizos, Max Welling, and Diederik P. Kingma. Learning sparse neural networks through l_0 regularization. *ICLR*, 2018. URL <https://arxiv.org/abs/1712.01312>.
- Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. *NeurIPS*, 2017. URL <https://arxiv.org/abs/1706.02275>.
- Chris J. Maddison, Andriy Mnih, and Yee Whye Teh. The concrete distribution: A continuous relaxation of discrete random variables. *ICLR*, 2017. URL <https://arxiv.org/abs/1611.00712>.
- André F. T. Martins and Ramón Fernandez Astudillo. From softmax to sparsemax: A sparse model of attention and multi-label classification. *ICML*, 2016. URL <https://arxiv.org/abs/1602.02068>.
- Georgios Papoudakis, Filippos Christianos, Lukas Schäfer, and Stefano V. Albrecht. Benchmarking multi-agent deep reinforcement learning algorithms in cooperative tasks. *NeurIPS Datasets and Benchmarks*, 2021. URL <https://arxiv.org/abs/2006.07869>.
- Ben Peters, Vlad Niculae, and André F. T. Martins. Sparse sequence-to-sequence models. *ACL*, 2019. URL <https://arxiv.org/abs/1905.05702>.
- Tabish Rashid, Mikayel Samvelyan, Christian Schroeder de Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Qmix: Monotonic value function factorisation for deep multi-agent rl. *ICML*, 2018. URL <https://arxiv.org/abs/1803.11485>.
- Tabish Rashid, Gregory Farquhar, Bei Peng, and Shimon Whiteson. Weighted qmix: Expanding monotonic value function factorisation. *NeurIPS*, 2020. URL <https://arxiv.org/abs/2006.10800>.
- Alexander Rutherford, Benjamin Ellis, Matteo Gallici, Jonathan Cook, Andrei Lupu, Gardar Ingvarsson, Timon Willi, Ravi Hammond, Akbir Khan, Christian Schroeder de Witt, Alexandra Souly, Saptarashmi Bandyopadhyay, Mikayel Samvelyan, Minqi Jiang, Robert Tjarko Lange, Shimon Whiteson, Bruno Lacerda, Nick Hawes, Tim Rocktaschel, Chris Lu, and Jakob Nicolaus Foerster. Jaxmarl: Multi-agent rl environments and algorithms in jax. *NeurIPS Datasets and Benchmarks*, 2023. URL <https://arxiv.org/abs/2311.10090>.
- Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Nantas Nardelli, Tim G. J. Rudner, Chia-Man Hung, Philip H. S. Torr, Jakob Foerster, and Shimon Whiteson. The starcraft multi-agent challenge. *AAMAS*, 2019. URL <https://arxiv.org/abs/1902.04043>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint*, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarsz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *ICLR*, 2017. URL <https://arxiv.org/abs/1701.06538>.

- Amanpreet Singh, Tushar Jain, and Sainbayar Sukhbaatar. Learning when to communicate at scale in multiagent cooperative and competitive tasks. *ICLR*, 2018. URL <https://arxiv.org/abs/1812.09755>.
- Kyunghwan Son, Daewoo Kim, Wan Ju Kang, David Earl Hostallero, and Yung Yi. Qtran: Learning to factorize with transformation for cooperative marl. *ICML*, 2019. URL <https://arxiv.org/abs/1905.05408>.
- Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by pid lagrangian methods. *ICML*, 2020. URL <https://arxiv.org/abs/2007.03964>.
- Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with back-propagation. *NeurIPS*, 2016. URL <https://arxiv.org/abs/1605.07736>.
- Peter Sunehag, Guy Lever, Audrunas Gruslys, Wojciech Marian Czarnecki, Vinicius Zambaldi, Max Jaderberg, Marc Lanctot, Nicolas Sonnerat, Joel Z. Leibo, Karl Tuyls, and Thore Graepel. Value-decomposition networks for cooperative multi-agent learning. *AAMAS*, 2017. URL <https://arxiv.org/abs/1706.05296>.
- J. K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Rodrigo Perez, Caroline Horsch, Clemens Dieffendahl, Niall L. Williams, Yashas Lokesh, and Praveen Ravi. Pettingzoo: Gym for multi-agent reinforcement learning. *NeurIPS 2021*, 2020. URL <https://arxiv.org/abs/2009.14471>.
- Muning Wen, Jakub Grudzien Kuba, Runji Lin, Weinan Zhang, Ying Wen, Jun Wang, and Yaodong Yang. Multi-agent reinforcement learning is a sequence modeling problem. *NeurIPS*, 2022. URL <https://arxiv.org/abs/2205.14953>.
- Yaodong Yang, Rui Luo, Minne Li, Ming Zhou, Weinan Zhang, and Jun Wang. Mean field multi-agent reinforcement learning. *ICML*, 2018. URL <https://arxiv.org/abs/1802.05438>.
- Yaodong Yang, Jianye Hao, Ben Liao, Kun Shao, Guangyong Chen, Wulong Liu, and Hongyao Tang. Qatten: A general framework for cooperative multiagent rl. *arXiv preprint*, 2020. URL <https://arxiv.org/abs/2002.03939>.
- Chao Yu, Akash Velu, Eugene Vinitzky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of ppo in cooperative, multi-agent games. *NeurIPS Datasets and Benchmarks*, 2022. URL <https://arxiv.org/abs/2103.01955>.
- Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. Big bird: Transformers for longer sequences. *NeurIPS*, 2020. URL <https://arxiv.org/abs/2007.14062>.